

Assessing the Ecological Impacts of Rain Gardens on Soil Health and Vital Ecosystem Functions.

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Seymour Library Rain Garden

Introduction

Rain gardens are engineered landscapes designed to capture and infiltrate stormwater while supporting ecosystem functions like nutrient cycling, organic matter accumulation, and microbial biodiversity (Sharma & Malaviya, 2021). Urban soils are often compacted and degraded from intensive management like mowing, while prairie ecosystems are seen as the ecological ideal with deep root structures that maintain soil health (Dutter et al., 2025, Pouyat et al., 2002).

This study compares soil health indicators across rain gardens, lawns, and a restored prairie to assess the ability of rain gardens to restore degraded soils.

Hypotheses: (1) Rain gardens will have better soil health than lawns; (2) Older rain gardens will show greater improvements than newer ones; (3) Prairie soils will have the highest indicators.

Methodology

Sampled 7 Sites at Knox College: 3 Rain Gardens (ages < 1 yr to ~ 10 yrs), 1 Restored Prairie (est. 2004), and 3 Lawns

Sampled April 2026, 3 cores per site (8-18 cm depths). Lab Analyses: bulk density, loss-of-ignition, soil texture, pH, nutrients (AgSource Labs), compaction.

Statistical analysis: One-way ANOVA with Tukey Kramer post-hoc ($\alpha = 0.05$).

Soil Quality Study Sites at Knox College

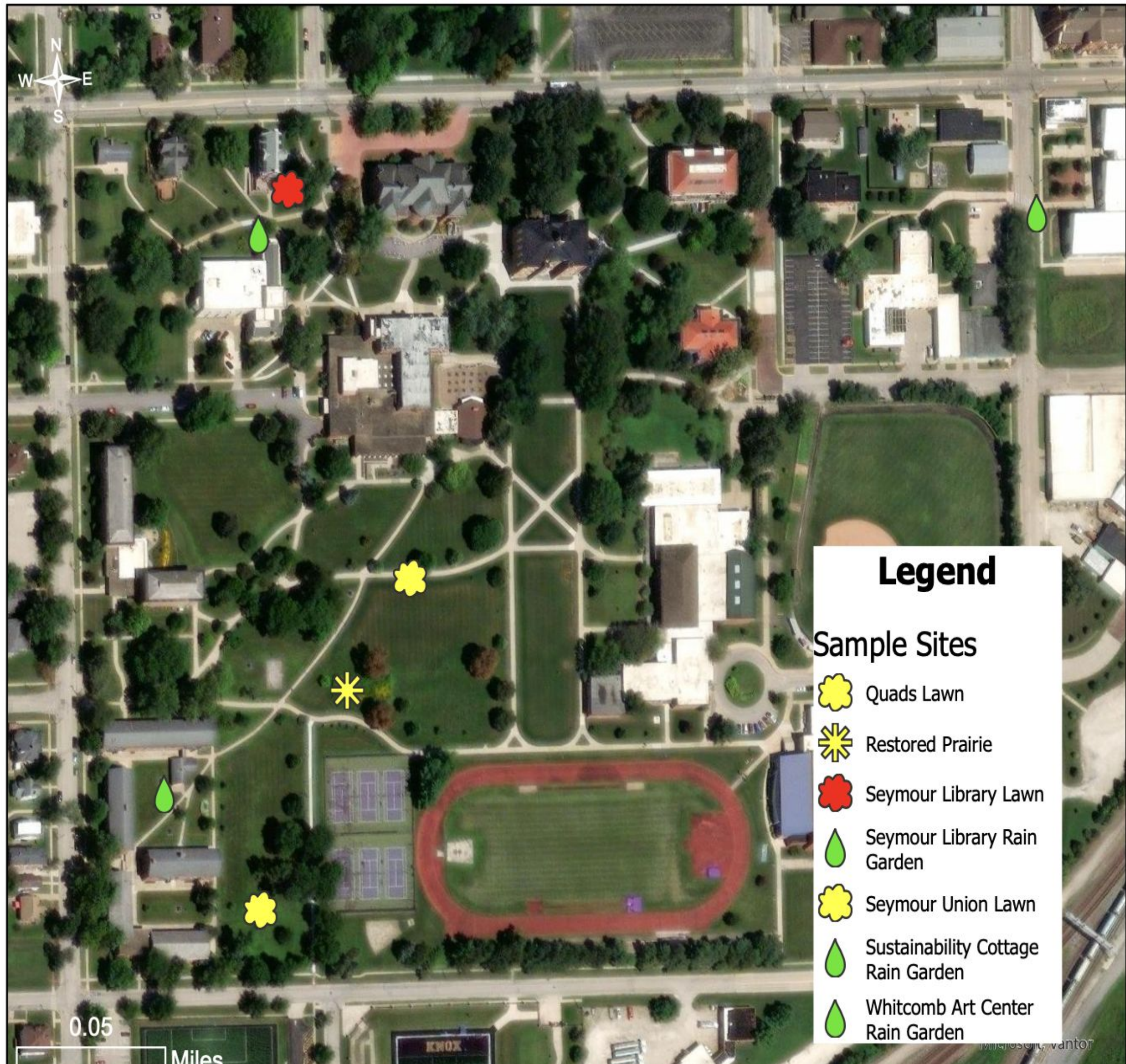


Figure 1. ArcGIS Pro map of study sites. Soil compaction levels across campus. tree = lawn, droplet = rain gardens, asterisk = prairie; green = < 150 psi, yellow = 150 - 250 psi, red = > 350 psi

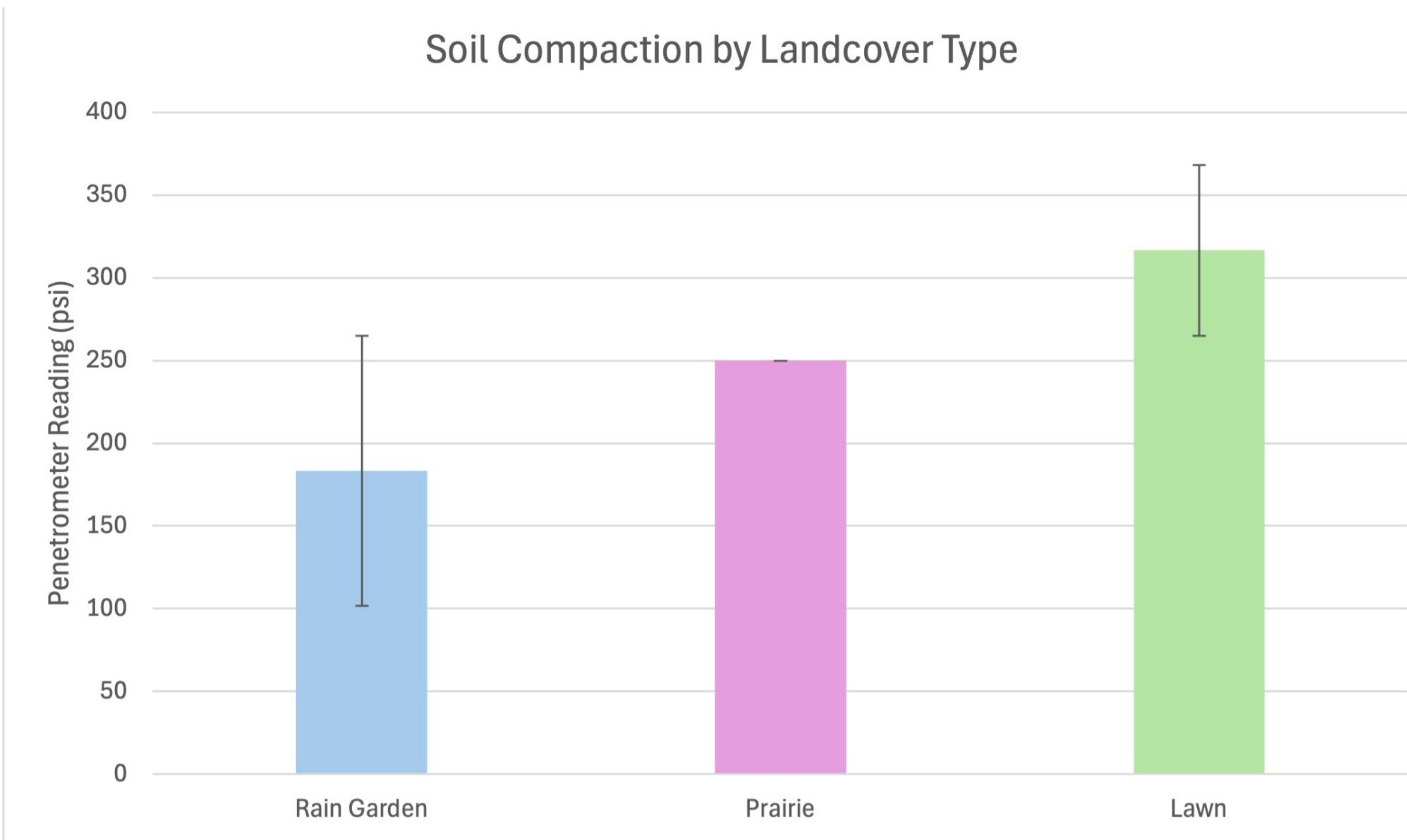


Figure 2. Mean soil compaction in psi across land cover types. Rain gardens had statistically significantly less compaction than lawns ($p = 0.015$).

Results & Discussion

Soil Compaction ($p = 0.015$): Rain gardens had significantly lower compaction (183 psi) than lawns (317 psi), supporting hypothesis 1. Native plants have deeper, more complex root systems that aerate soil and resist compaction better than mowed lawns.

Magnesium Content ($p = 0.005$): Prairie had the highest magnesium (445 ppm), likely because deeper roots access nutrient-rich soil horizons. Rain gardens had the lowest (171 ppm).

Age Effects: Established rain gardens had lower silt (42.7%) than new ones (56.8%, $p = 0.003$), and higher iron (27.8 vs 14 ppm, $p = 0.040$), partially supporting hypothesis 2.

Organic matter and other nutrients showed no significant differences. This was likely due to: (1) small sample size, which limits statistical power; (2) within-site variability that masked differences; (3) short timescale, chemical changes like organic matter accumulation happen slowly and may need decades to detect differences.

Conclusions

Rain gardens showed improved soil physical structure, suggesting that functional benefits for stormwater management can be realized relatively quickly. However, chemical improvements like organic matter accumulation take longer and weren't detected in this study. Therefore, rain gardens can deliver their primary benefit on a brief timescale.

Bibliography

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